

Computer Networks



Module 4: SNMP, ASN.1, Physical Layer

- SNMP
- ASN.1
- Physical Layer:
 - Data and signals
 - Digital transmission
 - Analog transmission
 - Bandwidth utilization
 - Transmission media

Bandwidth Utilization – Overview

Real communication links have **limited bandwidth**. Efficient use of available bandwidth is therefore essential.

Two broad approaches are used:

1. Multiplexing

Combines several signals or channels and transmits them simultaneously over a single high-bandwidth link.

2. Spreading

Expands the bandwidth of a signal and adds redundancy, mainly for **privacy and antijamming**.

- Multiplexing → mainly for **efficiency**.
- Spreading → mainly for **security and interference resistance**.

Multiplexing – Basic Concept

If the bandwidth of a link is greater than the bandwidth required by individual devices, the link can be shared.

Definition

Multiplexing is the simultaneous transmission of multiple signals over a single data link.

Instead of providing a separate link for every source, a **high-bandwidth link** can be shared.

Examples:

- Optical fiber
- Terrestrial microwave
- Satellite microwave

For n input lines:

n lines $\longrightarrow$ **MUX** $\longrightarrow$ one high-capacity link

one link $\longrightarrow$ **DEMUX** $\longrightarrow n$ output lines

Multiplexing – Link, Channel and Techniques

Link

The physical path used for communication.

Channel

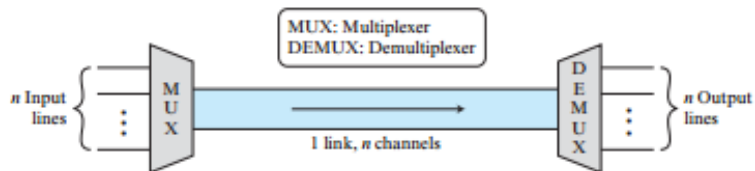
A portion of a link that carries a transmission between a particular pair of devices.

Basic Multiplexing Techniques

- **FDM** – Frequency Division Multiplexing
- **WDM** – Wavelength Division Multiplexing
- **TDM** – Time Division Multiplexing

Figure 7.44 – Multiplexed System

Figure 7.44 *Dividing a link into channels*



Frequency Division Multiplexing – FDM

FDM is an **analog multiplexing technique** used when the bandwidth of a link is greater than the combined bandwidths of the signals.

- 1 Each source produces a signal.
- 2 Each signal modulates a different carrier frequency.
- 3 The modulated signals are combined into one composite signal.
- 4 Guard bands separate adjacent channels.
- 5 At the receiver, filters separate the individual channels.

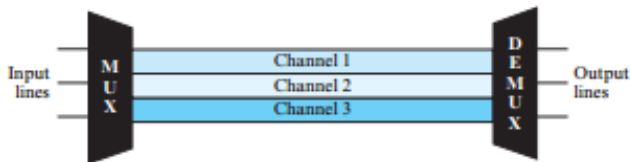
Important

The carrier frequencies must be selected so that the individual signals do not interfere with each other.

Digital sources can also be converted into analog signals before FDM.

Figure 7.45 – Frequency Division Multiplexing

Figure 7.45 *Frequency-division multiplexing*



Example 7.14 – FDM

A voice channel occupies **4 kHz**.

Three voice channels are combined using FDM.

$$3 \times 4 = 12 \text{ kHz}$$

Suppose the combined link occupies:



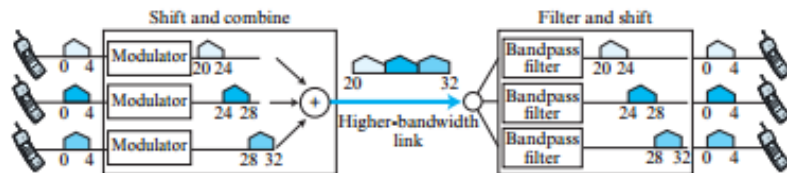
With no guard bands:

Channel	Frequency Range
1	20–24 kHz
2	24–28 kHz
3	28–32 kHz

At the receiver, filters select each 4-kHz range and the signals are shifted back to their original frequency range.

Figure 7.46 – Example 7.14

Figure 7.46 Example 7.14



Wavelength Division Multiplexing – WDM

WDM is designed for **fiber-optic communication**.

- Fiber-optic cables have extremely high bandwidth.
- A single optical signal may not use the entire capacity.
- Several optical signals can therefore share the same fiber.

Basic Idea

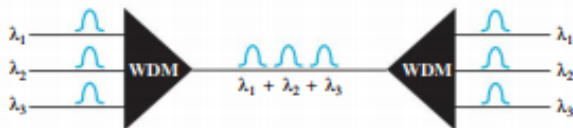
WDM is conceptually similar to FDM, but it operates using different **wavelengths/frequencies of light**.

- 1 Several narrow bands of light are generated.
- 2 Optical signals are combined into a wider optical band.
- 3 The combined signal travels through one fiber.
- 4 A demultiplexer separates the wavelengths at the receiver.

Application: SONET and high-capacity optical networks.

Figure 7.47 – Wavelength Division Multiplexing

Figure 7.47 Wavelength-division multiplexing



Time Division Multiplexing – TDM

TDM is a **digital multiplexing technique**.

Instead of dividing the link by frequency, different connections share the link at different **times**.

Source 1 → Source 2 → Source 3 → Source 4 → Source 1 → ...

- Each signal receives a time portion of the link.
- Portions are transmitted sequentially.
- TDM is multiplexing, not switching.
- Each source has a fixed destination in the multiplexed system.

Analog signals can also use TDM after sampling and conversion to digital form.

Two forms

Synchronous TDM

Statistical TDM

Figure 7.48 – Time Division Multiplexing

Figure 7.48 TDM



Synchronous TDM – Slots and Frames

In synchronous TDM, each input connection receives a fixed **time slot**, even when the source has no data.

For n input connections:

$$T_{\text{input}} = T$$


$$T_{\text{output slot}} = \frac{T}{n}$$

Therefore:

$$\text{Output data rate} = n \times \text{Input data rate}$$

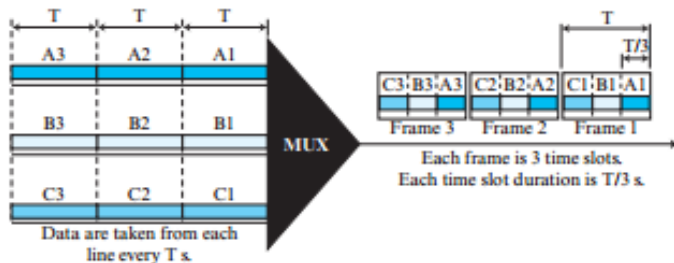
Frame

A complete cycle containing one slot for each input connection.

If each input unit takes T to transmit:

Figure 7.49 – Synchronous TDM

Figure 7.49 Synchronous time-division multiplexing



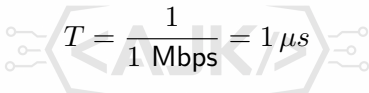
Example 7.15 – Synchronous TDM

Four input streams each operate at:

1 Mbps

Each input unit contains one bit.

Input bit duration:


$$T = \frac{1}{1 \text{ Mbps}} = 1 \mu s$$

For four inputs:

$$T_{\text{output bit}} = \frac{1}{4} \mu s = 0.25 \mu s$$

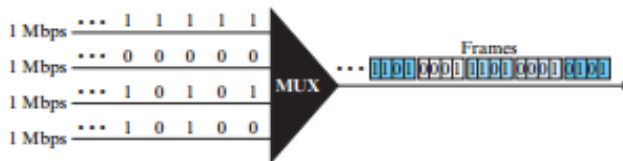
Therefore: Output rate = 4 Mbps
Frame contains: 4 bits/frame
Frame rate:

$1,000,000 \text{ frames/s}$

The frame rate is the same as the rate of each input stream.

Figure 7.50 – Example 7.15

Figure 7.50 Example 7.15



Example 7.16 – Digital Signal Hierarchy

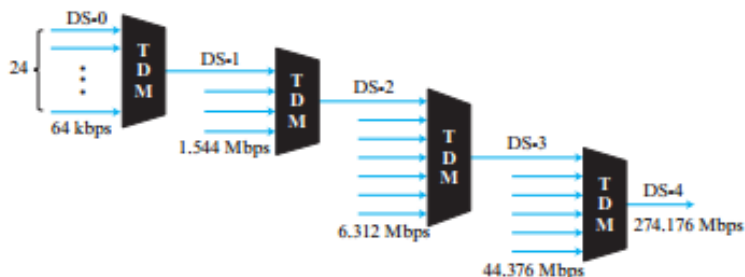
Level	Rate	DS-0 Equivalents	Hierarchy
DS-0	64 kbps	1	–
DS-1	1.544 Mbps	24	T-1
DS-2	6.312 Mbps	96	4 DS-1
DS-3	44.376 Mbps	672	7 DS-2 / T-3
DS-4	274.176 Mbps	4032	6 DS-3

Exact rates and overheads:

- $DS-0 = 64 \text{ kbps}$
- $DS-1 = 1.544 \text{ Mbps} = 24(64) \text{ kbps} + 8 \text{ kbps overhead}$
- $DS-2 = 6.312 \text{ Mbps} = 96(64) \text{ kbps} + 168 \text{ kbps overhead}$
- $DS-3 = 44.376 \text{ Mbps} = 672(64) \text{ kbps} + 1.368 \text{ Mbps overhead}$
- $DS-4 = 274.176 \text{ Mbps} = 4032(64) \text{ kbps} + 16.128 \text{ Mbps overhead}$

Figure 7.51 – Digital Signal Hierarchy

Figure 7.51 *Digital hierarchy*



Synchronous TDM can be inefficient when some input lines are idle.

Statistical TDM

Time slots are allocated dynamically only to input lines that have data to transmit.

- Number of slots is usually less than the number of input lines.
- The multiplexer checks input lines in round-robin order.
- Idle input lines are skipped.
- A statistical slot contains both data and destination address.

Why an address?

In synchronous TDM, fixed relationships and synchronization identify the destination. Statistical TDM dynamically assigns slots, so an address is required.

For N output lines:

$$n = \log_2 N$$

address bits are required. Example:

$$N = 8 \Rightarrow n = \log_2 8 = 3 \text{ bits}$$

Synchronous TDM vs. Statistical TDM

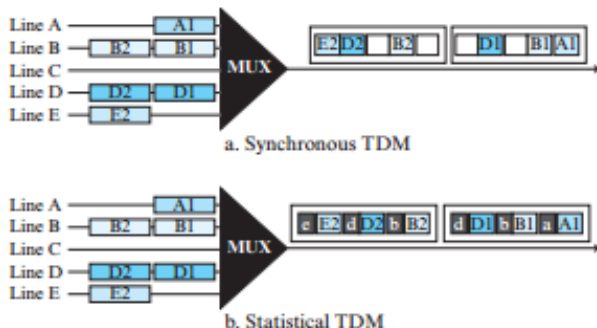
Synchronous TDM	Statistical TDM
Fixed time slots	Dynamic time slots
Every input receives a slot	Only active inputs receive slots
Idle slots may occur	Idle slots are avoided
Number of slots generally equals number of inputs	Number of slots can be smaller
No destination address required	Destination address required
Simple fixed mapping	Dynamic allocation

Main idea

Synchronous TDM provides predictable timing, while statistical TDM uses bandwidth more efficiently when input lines are intermittent.

Figure 7.52 – Statistical TDM

Figure 7.52 TDM slot comparison



Spread Spectrum – Concept and Purpose

Multiplexing combines several signals mainly for **efficiency**.

Spread spectrum expands the signal bandwidth mainly to provide:

- Privacy
- Antijamming capability
- Resistance to interference

It is particularly useful in:

- Wireless LANs
- Wireless WANs
- Shared wireless environments

If the original signal requires bandwidth B , spreading produces:

$$B_{ss} \gg B$$

Additional redundancy is introduced while spreading the original spectrum.

Spread Spectrum – Two Principles

Spread spectrum follows two important principles:

- 1 The allocated bandwidth is much larger than the bandwidth required by the original signal.
- 2 The expansion from B to B_{ss} is performed by a process independent of the original signal.

Spreading Code

A pseudorandom-looking pattern used to spread the original signal.

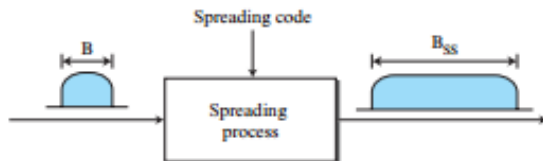
The receiver must know the appropriate spreading information to recover the original signal.

Two Techniques

- FHSS – Frequency Hopping Spread Spectrum
- DSSS – Direct Sequence Spread Spectrum

Figure 7.53 – Spread Spectrum

Figure 7.53 *Spread spectrum*



Frequency Hopping Spread Spectrum – FHSS

FHSS uses a set of carrier frequencies.

- There are M available carrier frequencies.
- Only one frequency is used at a time.
- The carrier changes periodically.
- The change occurs after every hopping period T_h .

The hopping sequence is controlled by a pseudorandom code.

$$B_{FHSS} \gg B$$

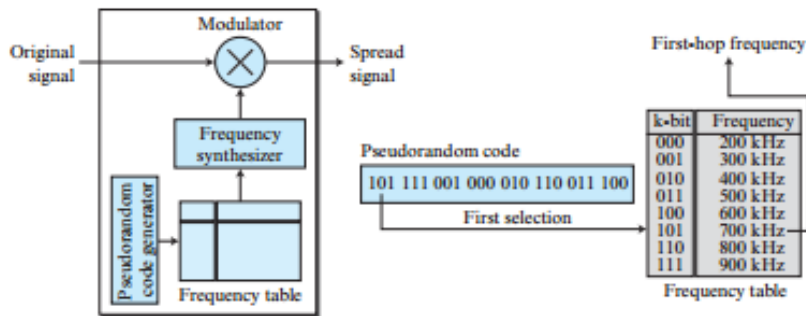
Basic Operation

PN generator $\rightarrow$ frequency selection $\rightarrow$ frequency synthesizer $\rightarrow$ transmitted signal

The receiver uses the same hopping sequence to follow the carrier.

Figure 7.54 – FHSS

Figure 7.54 Frequency hopping spread spectrum (FHSS)



FHSS – PN Generator and Frequency Selection

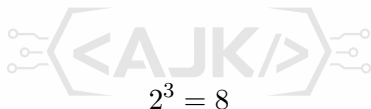
A PN (pseudorandom) code generator produces a k -bit pattern during each hopping period.

The pattern is used to select a frequency from a frequency table.

For:

$$M = 8, \quad k = 3$$

there are:



possible patterns.

Example hopping sequence:

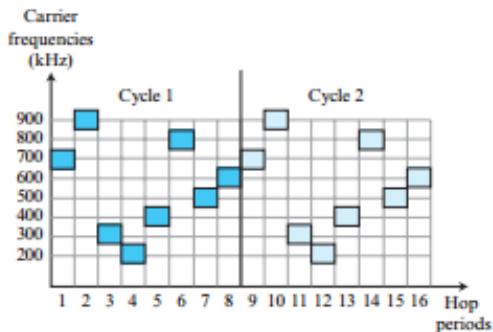
101, 111, 001, 000, 010, 011, 100, ...

Example frequency mapping:

101 $\rightarrow$ 700 kHz

Figure 7.55 – FHSS Frequency Hopping Example

Figure 7.55 FHSS cycles



Privacy:

An intruder who does not know the hopping sequence can access only a small portion of the transmission.

Antijamming:

A jammer may interfere with one hopping period, but the signal moves to another frequency during the next period.

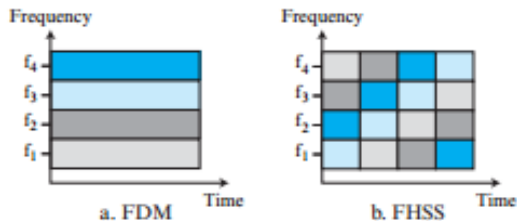
Bandwidth sharing:

If M hopping frequencies are available, multiple stations can share the same overall spread bandwidth.

- One station uses one frequency during a hopping period.
- The frequency allocation changes from hop to hop.
- Conceptually similar to FDM.

Figure 7.56 – FDM and FHSS

Figure 7.56 *Bandwidth sharing*



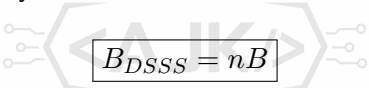
Direct Sequence Spread Spectrum – DSSS

DSSS expands bandwidth by replacing each data bit with a sequence of n smaller units called **chips**.

If original data rate is N :

$$\text{Chip rate} = nN$$

Therefore, approximately:


$$B_{DSSS} = nB$$

for the corresponding spreading factor.

Example – 11-chip Barker sequence

For $n = 11$, each data bit is represented by 11 chips.

Original rate: N

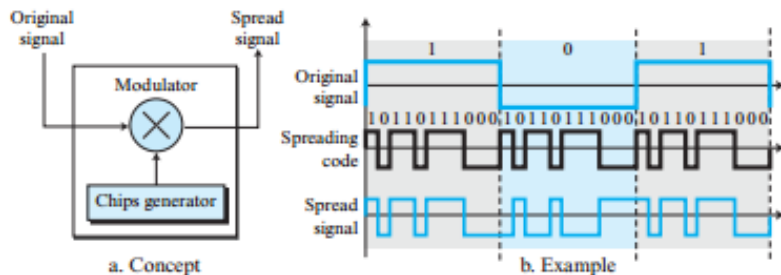
Spread rate: $11N$

Thus the bandwidth is approximately 11 times the original bandwidth.

Different stations can use different spreading codes.

Figure 7.57 – DSSS

Figure 7.57 DSSS



Bandwidth Utilization – Complete Summary

Multiplexing

Combines multiple signals over one physical link.

FDM	Frequency-based sharing; analog signals
WDM	Optical wavelength-based sharing
TDM	Time-based sharing; digital principle

TDM

Synchronous: fixed slots, possible idle slots.

Statistical: dynamic slots, address included.

Spread Spectrum

Expands bandwidth using a spreading code.

FHSS: carrier frequency changes with time.

DSSS: each bit is replaced by multiple chips.

Multiplexing → Efficiency
Spread Spectrum → Privacy + Antijamming